KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4109
Artificial Intelligence

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is artificial intelligence? To what extent are the following computer systems instances of artificial intelligence? Explain. (12)
   i) Supermarket bar-code scanners.
   ii) Web search engines.
   iii) Voice-activated telephone menus.
   iv) Internet routing algorithms.

   b) Consider a rabbit grazing in a carrot field. What are its sensors, actuators and environment? Discuss how its sensors and actuators are well suited to its environment? (13)

   c) Give a demystified classification of artificial intelligence. "The most subfield in AI focuses on smaller components thought to be necessary for producing intelligent programs" – Justify the statement. (10)

2. a) Explain why problem formulation must follow goal formulation. (08)

   b) What is PEAS? For each of the following activities, give a PEAS description of the task environment and characterize it in terms of the properties of (12)
   i) Bidding on an item at an auction, and
   ii) Automated taxi driving.

   c) How does a simple reflex agent differ from a goal based agent? "Learning agent is suitable for dynamic environment" – Justify this statement using an appropriate example. (15)

3. a) What is a fuzzy logic system? Explain. (10)

   b) Explain different operations on fuzzy sets. Use pictorial view for their clarity. (10)

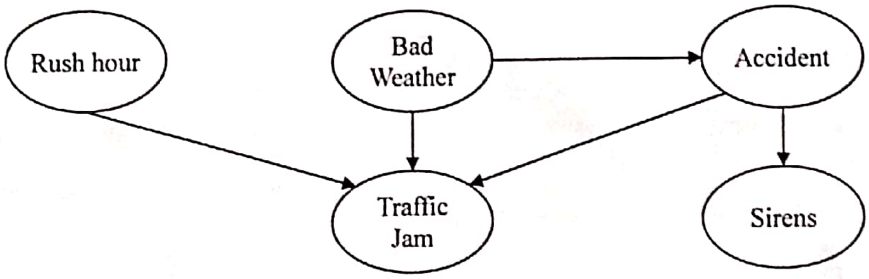
   c) Design a fuzzy logic based washing machine system using fuzzy expert system development methodology. (15)

4. a) Formulate graph coloring problem as a constraint satisfaction problem. Draw the constraint graph for an arbitrary graph. (11)

   b) What is a parse tree in NLP and for what purpose is it used? (05)

   c) Construct a grammar for the following sentence: (08)
   "The traveler shot the tiger with the gun" and also draw the syntactic tree.

   d) Derive the joint probability function for the following Bayesian network. (11)



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) How can you decide whether a system is an intelligent or not? Write down some capabilities of Artificial Intelligent (AI) system? (08)

   b) "The person is a toddler; If the person is a toddler, then the person is a child; If the person is a child and male, then the person is a boy; If the person is an infant, then the person is a child; if the person is a child and female, then the person is a girl; The person is a female;". Then express the above sentences into propositional logic and prove that the person is a girl. (12)

c) Consider the Wumpus world problem in the following 4×4 grid. The A, W, S, B, G and P in the grid define the facts of Wumpus world. (15)

4	S		B	P
3	W	B S G	P	B
2	S		B	
1	A Start	B	P	B
	1	2	3	4

Derive some rules from the Wumpus world problem using the grid and prove that there is a Wumpus ($W_{1,3}$) in the (1, 3) position.

6. a) Write down the limitations of propositional logic. (05)
   b) Transfer the following implicative clause into Conjunctive Normal Form (CNF): (08)

$$(Z \rightarrow Y) \rightarrow (\sim X \rightarrow (W \land P))$$

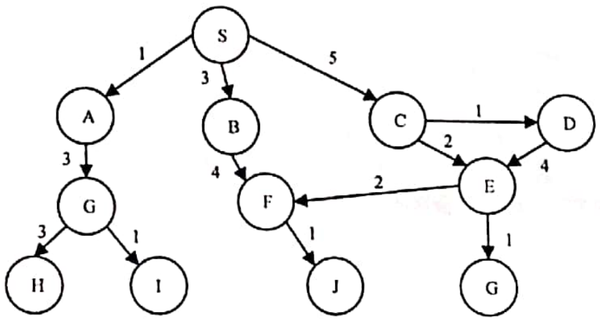
   c) Consider the following board of tic-tac-toe game of computer (O) versus human (×) and it's computer's turn to move. Then, how does computer take the better decision to move? Discuss with the heuristic function, the computer should consider. (12)

	O	
	×	O
×		

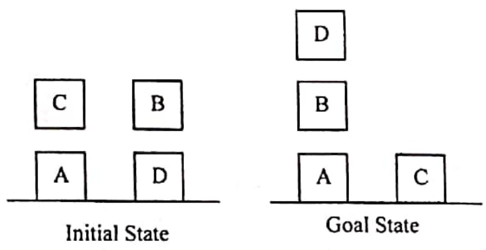
   d) Discuss the pruning conditions in alpha-beta pruning with proper example (10)

7. a) Write down the properties of quantifiers used in First-Order Logic (FOL). (05)
   b) Discuss the common mistakes of universal and existential quantifier of FOL with proper example. (12)
   c) Represent the following sentences into FOL. (18)
      i) Every house is a physical object.
      ii) Some physical objects are houses.
      iii) Every house is owned by somebody.
      iv) Everybody owns a house.
      v) Sue owns a house.
      vi) Peter does not own a house.

8. a) Find the path for destination node $\{J\}$ starting from node $\{S\}$ using uniform cost search strategy of the following graph. (12)



   b) Which search strategy is better between the A* search and greedy best first search and why? (10)
   c) A planning problem in the Block's world where a robot with one hand wants to rearrange the block from the initial state to the goal state. Now, design the actions and show the sequence of actions should perform by the robot using forward planning. (13)



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

B.Sc. Engineering 4th Year 1st Term Examination, 2020

Department of Computer Science and Engineering

CSE 4109

Artificial Intelligence

TIME: 1.5 hours FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) What is an agent? Explain the structure of a learning agent. (10)
   b) What is the significance of Turing test in artificial intelligence? Explain briefly. (10)
   c) What is PEAS? For each of the following activities, give a PEAS description of the task environment. (i) Playing soccer and (ii) Knitting a sweater. (10)

2. a) Develop a general structure of a fuzzy expert system. Hence, explain this system using "Air Conditional Control" problem. (12)
   b) "A* search is optimal if *h(n)* is consistent"-Justify the statement. (09)
   c) A doctor knows that the disease meningitis causes the patient to have stiff neck 70% of the time. The prior probability that a patient has meningitis is 1/50,000 and for stiff neck is 1%. Calculate, *P*(meningitis | stiff neck). (09)

3. a) What do you mean by Adversarial search? "Game playing is one kind of adversarial search"- Justify the statement. (08)
   b) What is local search? Explain how to use local search with min-conflict heuristic algorithm for solving n-queen problem. (10)
   c) Suppose one of the states of "Tic-Tac-Toe" is given in the following figure where you mark crosses (×'s) and machine marks circles (O's). Also suppose that now it is your turn to move. What will be your move using min-max procedure? Draw the complete search tree to show how you will find the wining state. Consider you as "max" and computer as "min". (12)

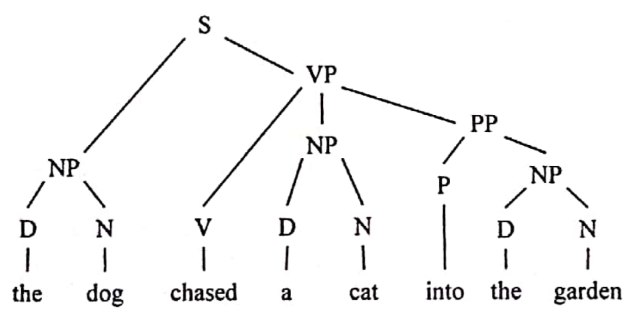
	O	O
	×	
×		

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) How can you construct a knowledge-based agent using declarative approach? Use example(s). (08)
   b) What is unification in First-Order Logic (FOL)? Explain different inference rule for FOL. (09)
   c) Use forward and backward chaining mechanism to prove that curiosity killed the cat for the following knowledge domain:
   "Everyone who loves all animals is loved by someone. Anyone who kills an animal is loved by none. Jack loves all animals, Either Jack or curiosity killed the cat, who is named Tuna" (13)

5. a) Do the label bracket of the following tree. (10)

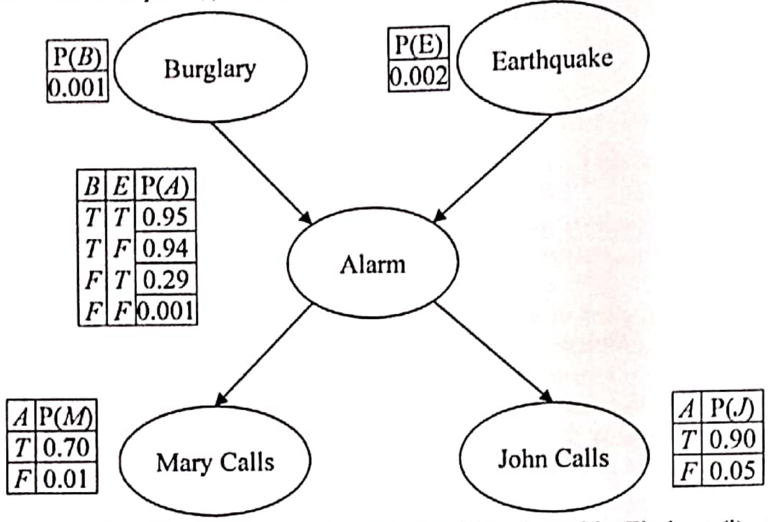


b) List out the several approaches for handling uncertainty. Briefly discuss the default reasoning approach. (12)

c) Define planning. Write down the goals of planning. (08)

6. a) Define syntax, semantics, pragmatics and discourse with example. (08)

b) Define probabilistic reasoning. Consider the following Bayesian Network for the following Alarm domain. Compute (i) $P(J,M,A,\neg B,\neg E)$; (ii) $P(J,M,A,E,B)$. (12)



c) Consider the probability distribution shown in the following table. Find out (i) $p(\neg\text{toothache})$; (ii) $p(\text{cavity} \wedge \text{toothache})$; (iii) $p(\text{toothache} \mid \text{cavity})$; (iv) $p(\neg\text{cavity} \mid \text{toothache})$. (10)

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

B.Sc. Engineering 4th Year 1st Term Examination, 2019

Department of Computer Science and Engineering

CSE 4109

Artificial Intelligence

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

SECTION A

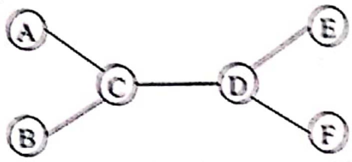
(Answer **ANY THREE** questions from this section in Script A)

1. a) How can you test whether a machine has reached the general intelligence level of human being? Explain it. (12)

b) What is an intelligent agent? Explain the learning agent model elaborately. (13)

c) What is PEAS? For each of the following activities, give a PEAS description of the task environment: (i) Automated Car Driving, (ii) Automated Robot Cleaner. (10)

2. a) What is CSP? Solve the following tree structured CSP where there are only three colors (Red, Green and Blue) available. (10)



b) Develop a local search algorithm for solving N-Queen problem using min-conflict heuristic. (12)

c) Suppose one of the states of "Tic-Tac-Toe" given in the following figure where you mark crosses (X's) and the machine marks circles (O's). Also suppose that now it is your turn to move. What will be your move using Mini-Max procedure? Draw the complete search tree to show how you will find the state. (13)

O	O	
	X	
		X

3. a) What is a fuzzy logic system? Explain clearly. (10)

b) Discuss different operations on fuzzy sets. Use pictorial view for their clarity. (10)

c) Develop a general structure of a fuzzy expert system. Hence, explain this system using a fuzzy logic controller of an A/C system. (15)

4. a) What is unification in the FOL? Why is it a key component in first-order inference algorithm? (07)

b) Using forward chaining and backward chaining, prove that Col. West is a criminal for the proposition: (15)

"The law says that it is a crime for an American to sell weapons to hostile nations. The country now, an enemy of America, has some missiles, and all of its missiles were sold to it by Col. West, who is an American".

c) Convert the logic $A \Leftrightarrow (B \vee C)$, into CNF. (08)

d) What are the advantages of FOL over propositional logic? (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

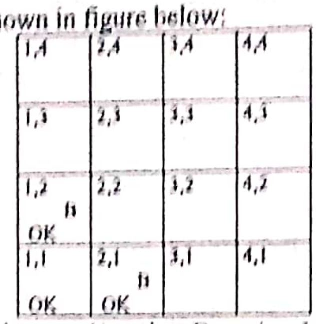
5. a) What are the differences between Exact Inference and Approximate Inference? Which techniques are used for Approximate Inference? (05)

b) What is Marginal independence? Calculate the probability values: (i) $p(\text{cavity} \vee \text{toothache})$, (ii) $p(\text{cavity})$, (iii) $p(\text{cavity} \mid \text{toothache})$, (iv) $p(\neg\text{cavity} \mid \text{toothache})$ using the following table. (05)

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.016	0.064	0.144	0.576

c) You have a new burglar alarm installed at home. It is fairly reliable at detecting burglary, but also responds on occasion to minor earthquakes. You also have two neighbours, John and Mary, who have promised to call you at work when they hear the alarm. Consider, Burglary=B, Earthquake=E, John=J, Mary=M and Alarm=A. $p(B)=0.001$, $p(E)=0.002$, $p(A \mid B,E)=\langle 0.95, 0.94, 0.29, 0.001 \rangle$, $p(J \mid A)=\langle 0.90, 0.05 \rangle$ and $p(M \mid A)=\langle 0.70, 0.10 \rangle$. Draw the Bayesian Network and calculate $p(\neg j, \neg m, a, b, e)$ and $p(j, m, a, \neg e, \neg b)$. (10)

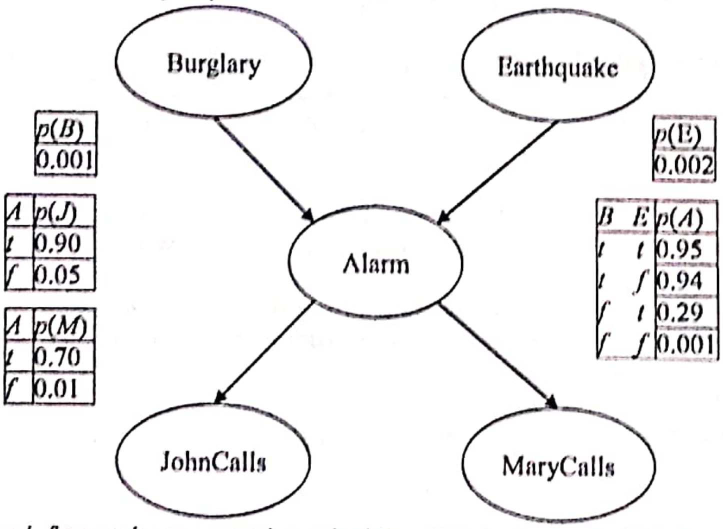
d) Consider the Wumpus world shown in figure below: (15)



Find out the value for $p(P_{1,3}\,|\,known, b)$ using Bayesian Inference. Show every consistent model for the frontier variables $P_{2,2}, P_{3,1}$.

6. a) What are the other approaches for uncertainty reasoning? Give a description on them. (12)
   b) What are the differences between typical Bayesian Net. and Dynamic Bayesian Net.? Draw the necessary figures. (06)
   c) Does the order of the Markov process has any impact on inference accuracy? Explain why or why not. (05)
   d) Using Hidden Markov Model derive the equations for the transition and the sensor probability for Robot Localization problem. (12)

7. a) What are the main differences between a probabilistic agent and logical agent? Define Prior and Posterior. (05)
   b) A doctor knows that the disease meningitis causes the patient to have stiff neck 70% of the time. The prior probability that a patient has meningitis is 1/50,000 and for stiff neck is 1%. Let, stiff neck=s and meningitis=m. Calculate $p(m\,|\,s)$. (05)
   c) Given, (10)
   $q_{cold} = p(\neg fever\,|\,cold, \neg flu, \neg malaria) = 0.6$; $q_{flu} = p(\neg fever\,|\,\neg cold, \neg flu, malaria) = 0.2$
   $q_{malaria} = p(\neg fever\,|\,\neg cold, \neg flu, malaria) = 0.1$.
   Build the conditional probability table using noisy-Or assumption.
   d) Consider the following Bayesian Network: (15)



By using inference by enumeration calculate $p(B\,|\,J, m)$ with proper diagram.

8. a) Prove that, $p(\neg A) = 1 - p(A)$ using axioms. (05)
   b) Suppose that, an online book retailer would like to provide overall evaluation of products based on recommendation from customers. The evaluation is a posterior distribution based on quality of the book. But the problem is, kind customers tend to give high recommendations while dishonest ones give very high or low. Briefly explain a Bayesian model for this real life scenario. (10)
   c) What is Dempster-Shaffer Theory? Explain with example. (06)
   d) Explain fuzzy sets and fuzzy logic with example. (06)
   e) Explain state model and transition model in Dynamic Bayesian Network with proper figures and examples. (08)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4109
Artificial Intelligence

TIME: 3 hours FULL MARKS: 210

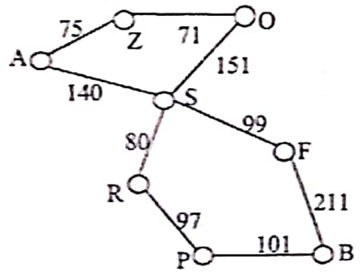
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define an Agent. Explain the structure of a learning agent. Use physical example(s). (12)
b) Describe the perception-action cycle of a robot which will move in a two dimensional grid world. (13)
c) "A* search is optimal if $h(n)$ is consistent". Justify the statement. (10)

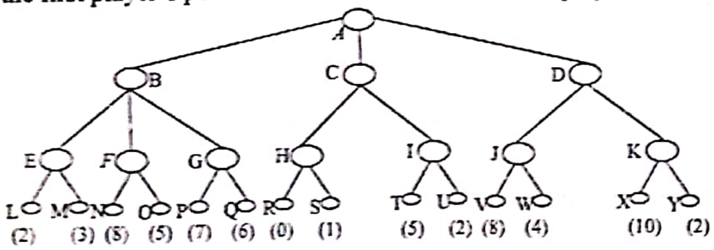
2. a) What is a heuristic function? Explain how to develop it for a particular problem. (10)
b) What is PEAS? For each of the following activities, give a PEAS description of the task environment: (i) Playing soccer and (ii) Knitting a sweater. (10)
c) Consider the following graph and associated information. Which one of greedy best and A* search will give better route from A to B? Explain using necessary details: (15)



Straight line distance to B

A	366
B	0
F	176
O	380
S	253
R	193
P	100
Z	374

3. a) How can you formally define a search problem? Explain using an example. (07)
b) "Uniform costs search act like BFS when step costs are equal"- justify. (05)
c) Prove each of the following statements or give a counter example: (10)
(i) "Iterative deepening search performs much worse than depth-first search".
(ii) "Uniform-cost search is a special case of A* search".
d) Consider the following game tree in which the static scores (in parentheses at tip nodes) are all from the first player's point of view. Assume that the first player is the maximizing player. (13)



(i) What move should the first player choose? (ii) What nodes would not need to be examined using alpha-beta algorithm, assuming that nodes are examined in left-to-right order?

4. a) What is a Constraint Satisfaction Problem (CSP)? Explain the different types of constraints those to be handled to solve CSPs. (08)
b) Use local search with min-conflict heuristic algorithm to solve n-queen problem. (09)
c) What is fuzzy set? Differentiate between fuzzy set and classical set. (06)
d) Based on the following "how to buy fresh fish" problem, construct a set of fuzzy rules: "If you go to the fish market to buy some fresh fish, you must know how to identify one. First, look at the eyes of the fish. If the eyes are reddish then the fish is no longer fresh, if the eyes are clear then the fish is fresh. Another tip is to look at fish's gills. The gills should be in rich red for the fish to be fresh. Otherwise, if the gills look pale red then it indicates that the fish is somewhat old and no longer fresh." Based on your given fuzzy rules, construct a Fuzzy Associative Memory (FAM) accordingly. (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What do you mean by resolution refutations? Explain using an example. (11)
b) What is uncertainty? How do you deal with uncertainty? Explain with example. (08)
c) Write down the differences between probabilistic and worst-case reasoning. (06)

d) Suppose you have a Wumpus world that consists of 2 cells: (1, 1) and (2, 1). In the truth table below, $P_{i,j}$ represents the fact that there is a pit at (i, j) and Knowledge Base (KB) represents the overall truth value of the KB.

$P_{1,1}$	$P_{2,1}$	KB
F	F	F
F	T	F
T	F	T
T	T	T

Does the KB $\models P_{2,1}$? Explain your answer. What does it mean for an inference procedure to be complete?

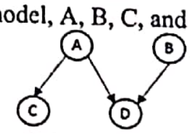
6. a) Consider the following table, where S = snow, R = rain and D = dry. (15)

	ϕ	{S}	{R}	{D}	{S, R}	{S, D}	{R, D}	{S, R, D}
Mfreeze	0	0.2	0.1	0.1	0.2	0.1	0.1	0.2
Mstorm	0	0.2	0.1	0.1	0.2	0.1	0.1	0.2
Mboth								

Here, assume two pieces of evidence: Temperature is below freezing and Barometric pressure is falling.
Compute (i) Mboth, (ii) Belief ({S, R}) using Mboth, (iii) Belief ({S, R}) using Mfreeze, and (iv) Belief ({S, R}) using Mstorm.

b) What is Dempster/Shaffer theory? Explain with example(s). (06)
c) Write down the differences between first order logic (FOL) and propositional logic. (05)
d) Translate the following sentences into FOL sentences: (09)
   i) Every member of the Hoofers club is either a skier or a mountain climber or both.
   ii) Some women are more knowledgeable than others except than herself.
   iii) Every fruit other than apricots and pineapples is bad and spoilt.
   iv) Apricots and pineapples are bad fruits and spoilt.

7. a) Define syntax, semantics, pragmatics and discourse with example. (09)
b) In the following graphical model, A, B, C, and D are binary random variables. (08)



How many parameters are needed to define the Conditional Probability Distributions (CPD's) for this Bayes Net?

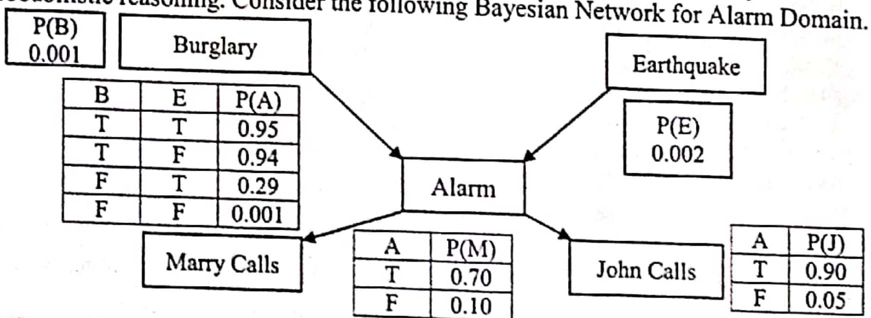
c) You will use the dataset below to learn a decision tree which predicts if people pass machine learning (yes or no), based on their previous GPA (High, Medium or Low) and whether or not they studied. (14)

GPA	Studied	Passed
L	F	F
L	T	T
M	F	F
M	T	T
H	F	T
H	T	T

For this problem, you can write your answer using $\log_2$, but it may be helpful to note that $\log_2 3 \approx 1.6$. (i) What is the entropy H(Passed)?, (ii) What is the entropy H(Passed | GPA)? and (iii) Draw the full decision tree that would be learned for this data set.

d) What is unification? Why is it so important to FOL? (04)

8. a) What is a parse tree in NLP, and for what is it used? (05)
b) Construct a Grammar for the following sentence: "The policeman shot the theif with the gun". Also draw the syntactic tree. (09)
c) What is rule based expert system? Explain basic structure of a rule based expert system. (08)
d) Define probabilistic reasoning. Consider the following Bayesian Network for Alarm Domain. (13)



B	E	P(A)
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.001

A	P(M)
T	0.70
F	0.10

A	P(J)
T	0.90
F	0.05

Compute (i) $P(\neg J, \neg M, B, E)$ and (ii) $P(J, M, A, \neg E, B)$

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

B.Sc. Engineering 4^{th} Year 1^{st} Term Examination, 2017

Department of Computer Science and Engineering

CSE 4109

Artificial Intelligence

TIME: 3 hours

FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) How can you test whether a computer has reached the general intelligence level of human being? Explain it. (12)
   b) What is an agent? Explain the perception-action cycle of a point robot agent wandering in a two-dimensional grid world. (13)
   c) What is PEAS? For each of the following activities, give a PEAS description of the task environment: (10)
      i) Automated car driving;
      ii) Medical diagnosis system.

2. a) What is a min-max procedure? Explain this procedure using "Tic-Tac-Toe" game (12)
   b) What makes an admissible heuristic? Why is this important for search? Explain why using the number of squares in the wrong position for the 8-puzzle is a bad idea even through it is admissible. What is a better heuristic? (14)
   c) Why problem formulation must follow goal formulation? Explain. (09)

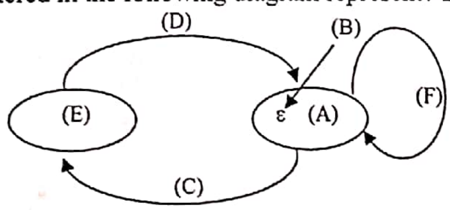
3. a) What is a fuzzy set? Differentiate between fuzzy and crisp sets using example. (10)
   b) Explain different fuzzy linguistic hedges using example(s). (10)
   c) Develop a general structure of a fuzzy expert system. Hence, explain this system using "Air conditioner control" problem. (15)

4. a) Formulate graph coloring problem as a constraint satisfaction problem. Draw the constraint graph for an arbitrary graph. (09)
   b) Define horizon effect. Is there any way to overcome the horizon effect? Explain. (07)
   c) What is the drawback of forward checking? How the arc consistency is used to solve this? (12)
   d) Prove that uniform costs search act like BFS when step costs are equal. (07)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is propositional logic? With respect to AI, what is it good for? (07)
   b) Define knowledge based agent. Write down the architecture of a knowledge based Agent. (10)
   c) What does each of the lettered in the following diagram represent? Explain. (08)

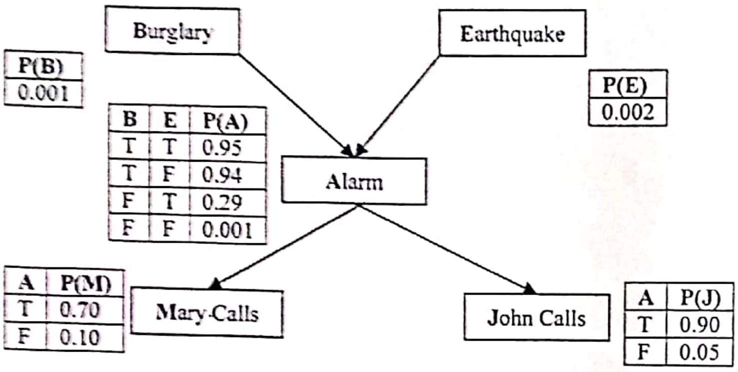


   d) Suppose you have a Wumpus world that consists of 2 cells: (1,1) and (2,1). In the truth table below, P_{ij} represents the fact that there is a Pit at (i, j) and KB represents the overall truth value of the knowledge base. (10)

$P_{1,1}$	$P_{2,1}$	KB
F	F	F
F	T	F
T	F	T
T	T	T

Does the knowledge Base$\models P_{2,1}$? Explain your answer. What does it mean for an inference procedure to be complete?

6. a) Define Probabilistic Reasoning. Consider the following Bayesian Network for Alarm Domain. (13)



Compute (i) $P(J, M, A, \neg B, \neg E)$ (ii) $P(J, M, A, E, B)$

b) What is uncertainty? What are the types of uncertainty? Explain with suitable example(s). (07)

c) List the approaches for Handling uncertainty. Explain one of them. (10)

d) Make a relationship between Problem Domain and knowledge Domain. (05)

7. a) What do you mean by NLP? Construct a grammar for the following sentence: "The fact that fishes swim astonished him". Also draw the syntactic tree. (15)

b) Define syntax, semantics, pragmatics and discourse with example. (12)

c) What do you mean by default logic? Give formal definition of default logic. (08)

8. a) What do you mean Dempster/Shaffer theory? Explain with example(s). (06)

b) Consider the following table, where S=snow, R=rain and D=Dry. (15)

	ϕ	{S}	{R}	{D}	{S, R}	{S, D}	{R, D}	{S, R, D}
Mfreeze	0	0.1	0.1	0.2	0.1	0.1	0.1	0.3
Mstrom	0	0.1	0.2	0.2	0.1	0.1	0.2	0.1
Mboth								

Here, assume two pieces of evidence: Temperature is below freezing and Barometric pressure is falling.

Compute (i) Mboth (ii) Belief ({S R}) using Mboth (iii) Belief ({S R}) using Mfreeze (iv) Belief ({S R}) using Mstorm.

c) What is rule based expert system? Explain basic structure of a rule based expert system. (09)

d) What is conflict resolution? Explain with example(s). (05)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2016
Department of Computer Science and Engineering
CSE 4109
Artificial Intelligence

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

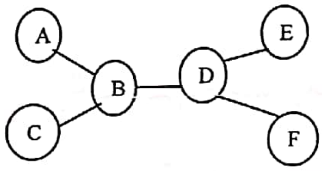
SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Artificial Intelligence? Explain, how can we test a machine has reached the general intelligence level of a human being. (12)
b) What is intelligent agent? Explain the perception-action cycles of a point robot wandering in a two dimensional grid-world. (13)
c) What is PEAS? For each of the following activities, give a PEAS description of the task environment: (i) Playing soccer; (ii) Knitting a sweater. (10)

2. a) Explain why problem formation must follow goal formation. (10)
b) Which of the followings are true and which are false? Explain your answers. (15)
(i) "Depth-first search always expands at least as many nodes as A* search with admissible heuristics".
(ii) "$h(n)=0$ is an admissible heuristic for 8-puzzle".
(iii) "Breadth-first search is complete even if step costs are allowed".
c) Prove each of the following statements, or give a counter example: (10)
(i) "Uniform-cost search is a special case of A* search".
(ii) "Iterative deepening search performs much worse than depth-first search".

3. a) What is Constrained Satisfaction Problem? Solve the following tree structured CSP. Consider there are only three colors (Red, Green, and Blue) available. (10)



b) Develop a local search algorithm for solving CSP using min-conflicts heuristic. (12)
c) Suppose one of the states of "Tic-Tac-Toe" given in the following figure where you mark crosses ($x's$) and the machine marks circles (O's). Also suppose that now it is your turn to move. What will be your move using mini-max procedure? Draw the complete search tree to show how you will find the state. (13)

	O	O
	×	
×		

Consider you as Max and the machine as Min.

4. a) What do you mean by Fuzzy System? Explain. (10)
b) Discuss the different operations on Fuzzy sets. Use pictorial view for their clarity. (10)
c) Develop a general structure of a fuzzy expert system. Hence, explain this system using a "Hotel Tipping" problem. (15)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define "entailment" and "model of a sentence" with proper example. (08)
b) Define model checking. What is the problem of using it? Explain. (10)
c) Express the followings sentences in propositional logic. (12)
"The person is a toddler; if the person is a toddler then the person is a child; if the person is a child and male then the person is a boy; if the person is an infant then the person is a child; if the person is a child and female then the person is a girl; the person is female." Also prove that the person is girl using rule of inferences.
d) Find the Conjunctive Normal form (CNF) for $p \leftrightarrow (\bar{p} \vee \bar{q})$. (05)

6. a) Find predicate, function and term from the following sentences: (06)
(i) Brother(John, Richard)
(ii) >(Length(LeftlegOf(Richrad)),(Length(LeftlegOf(John)))
(iii) Sibling(John, Richard)$\Rightarrow$ Sibling(Richard, John)

b) Consider the following sentences: (14)
"Tony, Mike, and John belong to the Alpine Club. Every member of the Alpine Club who is not a Skier is a mountain climber. Mountain climbers do not like rain, and anyone who does not like snow is not a skier. Mike dislikes whatever Tony likes, and likes whatever Tony dislikes. Tony likes rain and snow." Express the above sentences in the First Order Logic (FOL). Also prove that the sentences logically entail that there is a member of the Alpine Club who is mountain climber but not a skier using resolution proof.

c) From the following knowledge base (KB) find who is hard worker using answer-extraction process? (08)

KB

$\forall x GradStudent(x) \Rightarrow Student(x)$

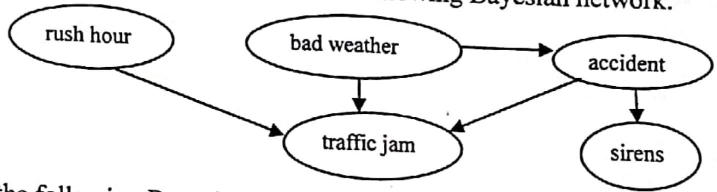
$\forall x Student(x) \Rightarrow HardWorker(x)$

$GradStudent(John)$

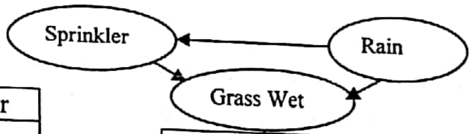
d) How does backward chain work? Explain. (07)

7. a) Consider you have four attributes A, B, C, D and 25 records. How will you construction a decision tree using the given attributes and records? Explain. (12)
b) Define pruning of a decision tree. Why is it necessary? Explain. (07)
c) What is default logic? Explain syntax of default logic with example. (10)
d) Differentiate Monotonic logic and Non-monotonic logic using example. (06)

8. a) Derive the joint probability function of the following Bayesian network. (08)



b) Consider the following Bayesian network. (12)



Rain
T | F

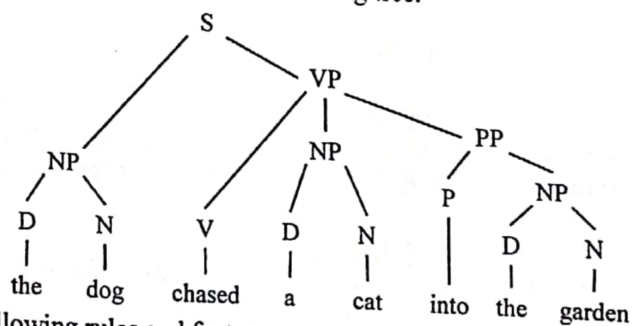
Rain T	Rain F
0.3	0.7

	Sprinkler	
Rain	T	F
F	0.3	0.7
T	0.03	0.97

Sprinkler	Rain	Grass Wet	
		T	F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.85	0.15
T	T	0.97	00.3

What is the probability that it is raining given that the grass is wet?

c) Define syntactic tree. Label bracket the following tree. (06)



d) Consider the following rules and facts: (09)

$S \Rightarrow [np, vp]$

$np \Rightarrow [pn]$

$vp \Rightarrow [iv]$

$vp \Rightarrow [tv, np]$

Rules

$lex(vincent, pn).$

$lex(mia, pn).$

$lex(died, iv)$

$lex(loved, tv).$

$lex(shot, tv).$

Facts

Using top down parsing show that mia loved vincent.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2015
Department of Computer Science and Engineering
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Artificial Intelligence

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
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SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Artificial Intelligence? How can we test whether a machine has reached the general intelligence level of a human being? Explain it. (15)
   b) What is perception-action cycle? Explain the perception-action cycle of a point-robot which will travel through a two-dimensional grid world. (12)
   c) What is an artificial agent? Explain the properties of an artificial agent. (08)

2. a) Explain uniform-cost search with example. (10)
   b) What is the problem of DFS strategy? How can the problem be solved? Explain it with an example. (13)
   c) "Greedy best-first search strategy suffers from the same defects as DFS strategy"-justify the statement. (12)

3. a) What is heuristic search? Explain using an example. (10)
   b) "A* search is optimal if h(n) is consistent"-justify the statement. (10)
   c) Define Constraint Satisfaction Problems. Explain the deferent types of Constraints those to be handled to solve CSPs. (10)
   d) Write down the local-search with min-conflict heuristic to solve the 8-Queen problem. (05)

4. a) What is adversarial search? Explain. (06)
   b) What is Min-Max procedure? Explain its working principle using a proper example. (12)
   c) What do you mean by fuzzy logic? Explain the concept of fuzzy logic using a physical example. (12)
   d) Why two player game playing a zero-sum action? Explain. (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define Validity, Entailment and Skolem constant with example. (09)
   b) Why is propositional logic considered as a weak knowledge representation language? (06)
   c) State the propositional inference rules. Consider the following statements:
   "Either cat fur or dog fur was found at the scene of the crime. If dog fur was found, officer Thompson had an allergy attack. If cat fur was found then Macavity is responsible for the crime. But officer Thompson didn't have an allergy attack". Using the inference rules show that Macavity is responsible for the crime. (12)
   d) What do you mean by horn clause and definite clause? Consider the following knowledge base $(\neg A \vee \neg B \vee C) \wedge \neg (A \forall \neg B)$. Is the knowledge base is horn form? Are all clauses definite clauses? (08)

6. a) How can you build a knowledge-based agent using declaratic approach? (06)
   b) Convert the logic $A \Leftrightarrow (B \vee C)$ into CNF. Distinguish between Horn clauses and definite clauses. (12)
   c) Consider the following knowledge base (12)

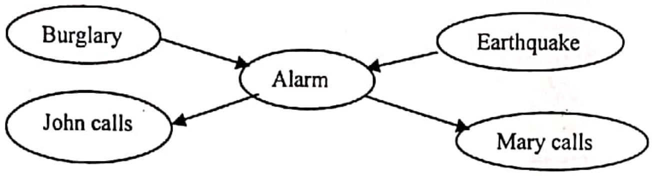
$$A \Leftrightarrow B$$
$$C \wedge D \Rightarrow A$$
$$Q \wedge C \Rightarrow D$$
$$P \wedge A \Rightarrow C$$
$$P \wedge Q \Rightarrow C$$
$$P$$
$$Q$$

   Draw corresponding AND-OR graph and using Backward chaining find out whether the query B is true.

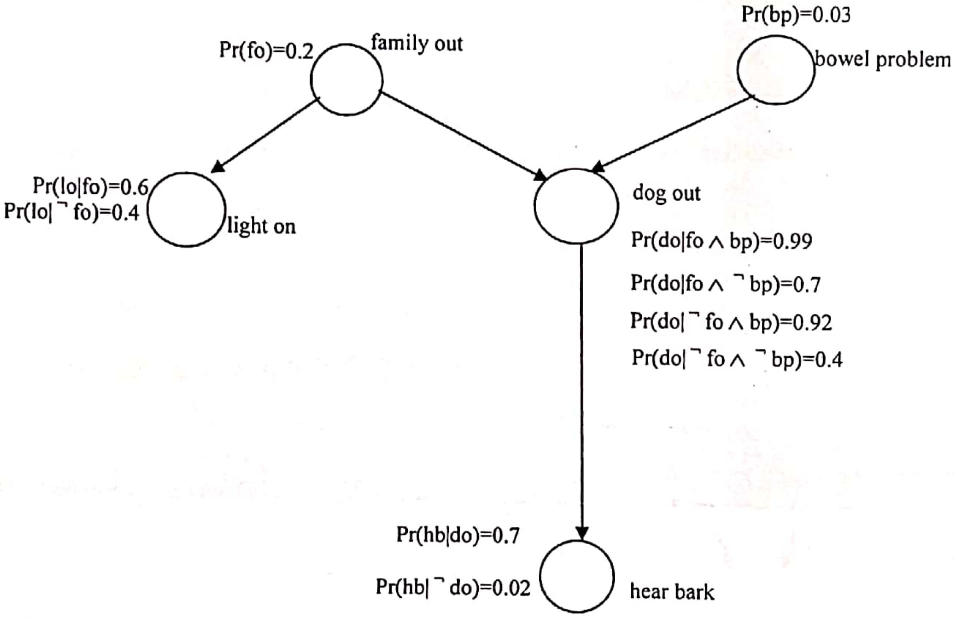
d) How can you define the relation of logical entailment between sentences? (05)

7. a) What is decision tree? Write down the basic idea of decision tree algorithm. (10)
   b) Construct a grammar for the following sentence. Also draw the syntactic tree. (17)
   "The fact that dogs bark shocked him". Use top down approach to parse it.
   c) What is pruning? What is the goal of pruning? (08)

8. a) Derive the product formula for Bayesian networks. Show the steps to calculate the product formula of the following Bayesian network. (12)



   b) Consider the following Bayesian network (17)



   Find out the joint probability distribution. Calculate the probability that the family is out, given that the light is on and we hear barking.
   c) What do you mean by anaphora and metaphore in Natural Language Processing (NLP)? (06)